

VISUAL WORLD PARADIGM



Do we predict next words the same as LLM?

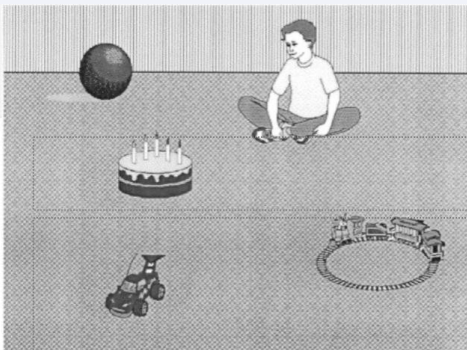
Acquisition and Analysis of Eye Tracking Data · University of Stuttgart · 2026

Are the prediction **probabilities of LLMs on subsequent references** similarly distributed to probabilities estimated with **human anticipatory eye movements** in **Visual World Paradigm**?

Visual World Paradigm

A visual world created by researchers to infer human language predictions by measuring where people look while listening to spoken language.

Altmann&Kamide(1999)



LLM prediction probabilities

Input to LLM

"The boy will eat the..."

Computes surprisal per token

$\text{surprisal} = -\log_2(P_{\text{token}})$ — direct output of `llm_generate.py` (Titus von der Malsburg)

Summed over each candidate object's tokens

$\text{surprisal}(\text{cake}), \text{surprisal}(\text{toy car}), \text{surprisal}(\text{ball})...$

Converted into a normalized probability

$P_{\text{norm}} = \text{softmax}(2^{\sim \text{surprisal}})$, sums to 1 across the scene's objects

Human anticipatory eye-movements

Hears the sentence ...

"The boy will eat the..."

Verb narrows the next possible word

restrictive verb → only one object still fits

Gaze drifts to the likely object

before the target word is even spoken — anticipatory, not reactive

Fixation proportion = the human signal

% looks to target vs. distractors in the critical window

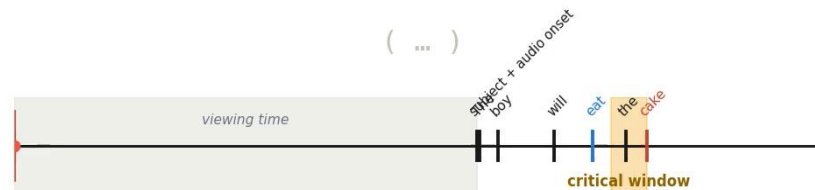
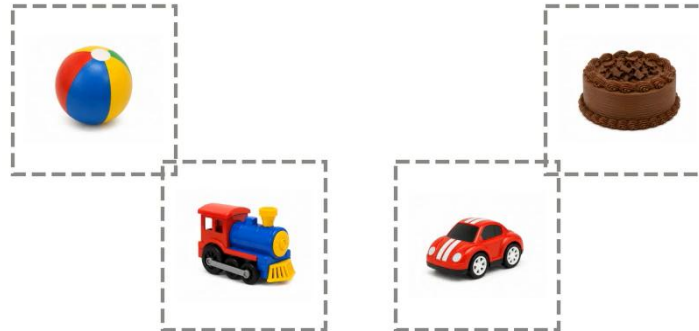
Experimental Design (~24 participants , 50 trials/p, ~15 mins/trail)

Trial Timeline



What the participant sees & hears — item 1
GPT-2 prediction (boxes) vs. real human gaze (red trail)

Viewing time — no subject yet



Implementation of Experiment and Problems & Fixes

Stimuli

Text

- Generation & check against original Altmann & Kamide items
- Kept subject consistent per item across both conditions

Audio

- Consistent speaking rate across items

Images

- Real photos only or cartoon
- Balanced visual salience (size, saturation) across objects in a scene

→ Fix

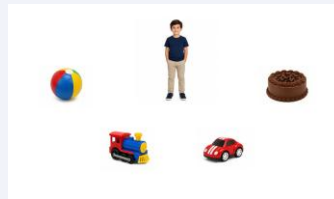
- Mimic the original experiment's materials/method wherever possible

Balance

Left-right visual bias

→ target position rotated across 4 physical slot

→ fixed semicircle layout



Verb-condition effect on results

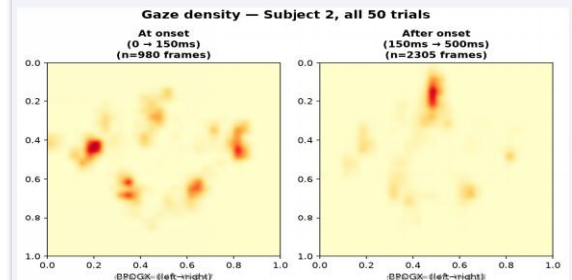
→ restrictive vs. non-restrictive per item, same target + distractors

Position	Restrictive	Non-restrictive
position1	Group 1	Group 2
position2	Group 3	Group 4
position3	Group 5	Group 6
position4	Group 7	Group 8

Trial Procedure & Open Issues

Procedure design

- Pre-trial object-preview window before audio starts, so layout is known before language begins
- Show the picture with non-subject first and then subject in the middle to center the gaze

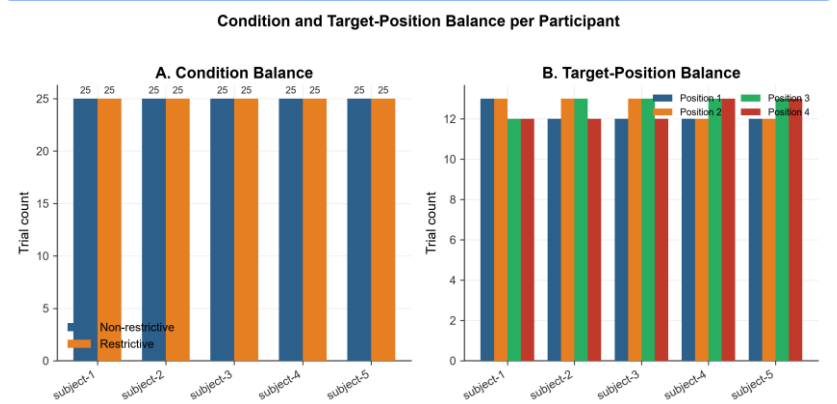


Quality Assurance – Design Check

Counterbalancing validity and known logging issues, n = 5 subjects

✓ Counterbalancing confirmed across all subjects; 2 known infrastructure issues persist — documented workarounds in place

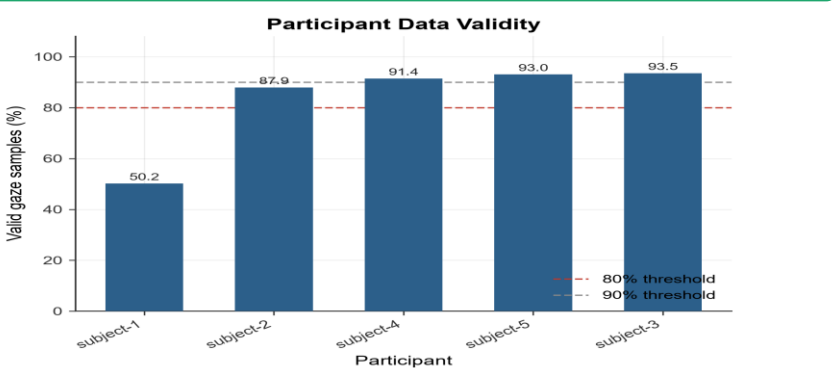
① Condition & position balance



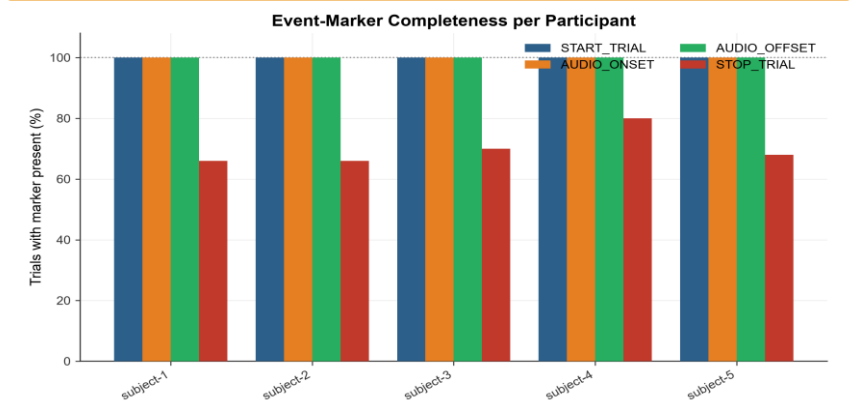
Every subject: exactly 25 restrictive / 25 non-restrictive, target position within 12–13 trials each.

✓ Latin-square design confirmed

3 Was enough valid data captured?



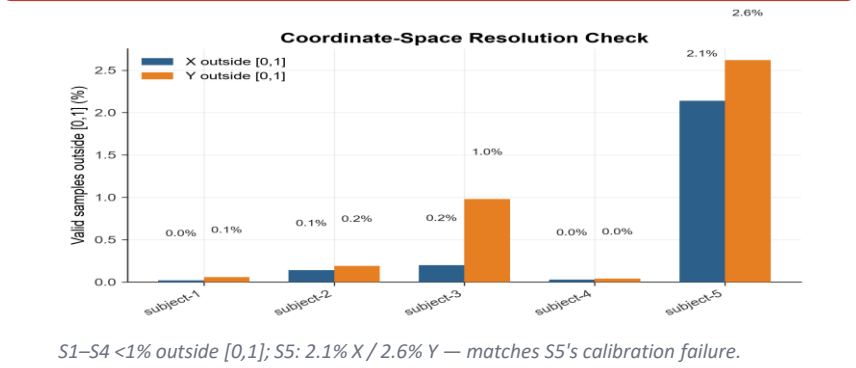
② Event-marker completeness



START_TRIAL / AUDIO_ONSET / AUDIO_OFFSET: 100% for all 5. STOP_TRIAL: 65–80% — known logging bug.

⚠ STOP_TRIAL bug persists (workaround: use AUDIO_OFFSET as trial-end marker)

④ Resolution / coordinate check



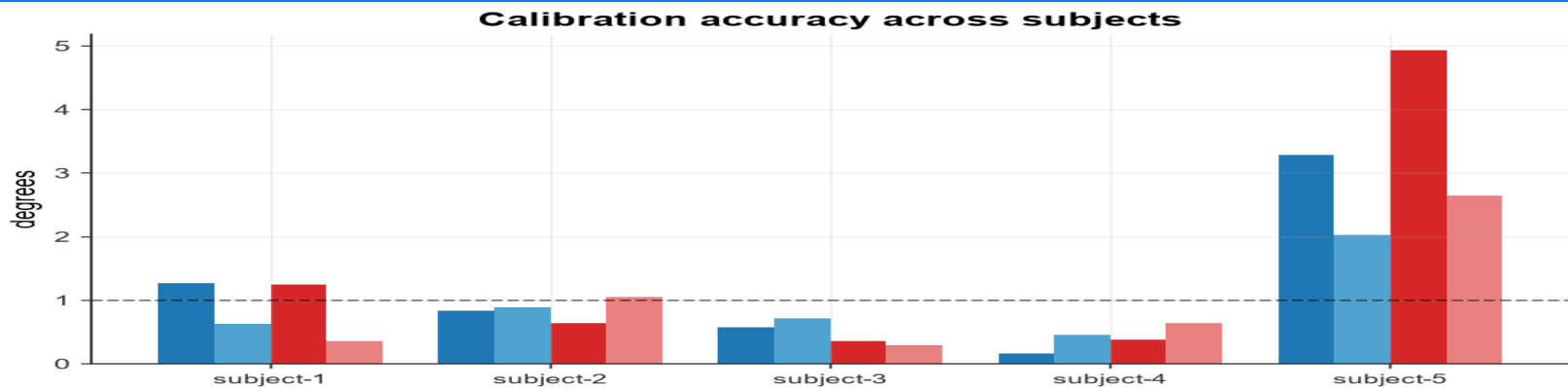
S1–S4 <1% outside [0,1]; S5: 2.1% X / 2.6% Y — matches S5's calibration failure.

Quality Assurance – Can we Trust the Data ?

4 independent checks. $n = 5$ subjects — each supported by a figure

⚠ 2 of 5 subjects flagged for data-quality issues — recommend exclusion or re-test (details below)

① Was the tracker set up correctly?



S3, S4 well under 1°. S1 (LX 1.27°, RX 1.25°) and S2 (RY 1.05°) marginally exceed. S5 fails badly on all 4 axes (up to 4.9°)

--- 1° threshold

■ LX

■ LY

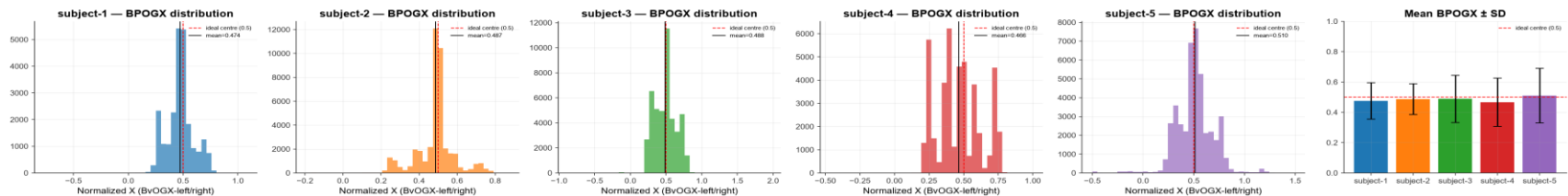
■ RX

■ RY

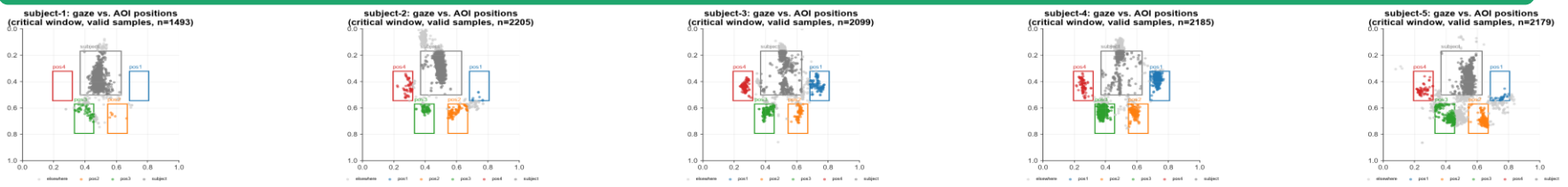
✗ S5: calibration failure
✗ S1: only half the session usable

③ Any systematic left/right gaze bias? — ✓ no bias in any subject

Horizontal Gaze Bias Check (BPOGX, valid samples only)



④ Does gaze land where the objects actually are? — ✓ AOI geometry confirmed



Analysis Pipeline

preliminary annotation & metadata



- per word-annotated audio files (ctc-forced aligner tool ^[1] & POS tagger ^[2])
- Subject-X.csv & subject-X.tsv & subject-3_log

preprocessing

- retrieve group and object-positions
- restrict samples to prediction window (Verb Onset → Noun Onset)
- remove invalid samples

Human probability estimation

Calculate the percentage per AOI
& normalize by total number



Map AOIs to Objects

AOI1-‘cake’
AOI2-‘book’
AOI3-‘ball’
AOI4-‘car’

AOI1 6

AOI2 0

AOI3 8

AOI4 0

[1]-<https://github.com/MahmoudAshraf97/ctc-forced-aligner>

[2]-<https://github.com/explosion/spaCy>

Analysis Pipeline II

Aggregate Across participants

Average over all participants' distributions, per sentence



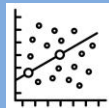
$$P_{\text{Human}}(o) = (1/N) \cdot \sum P_i(o)$$

LLM probability distribution

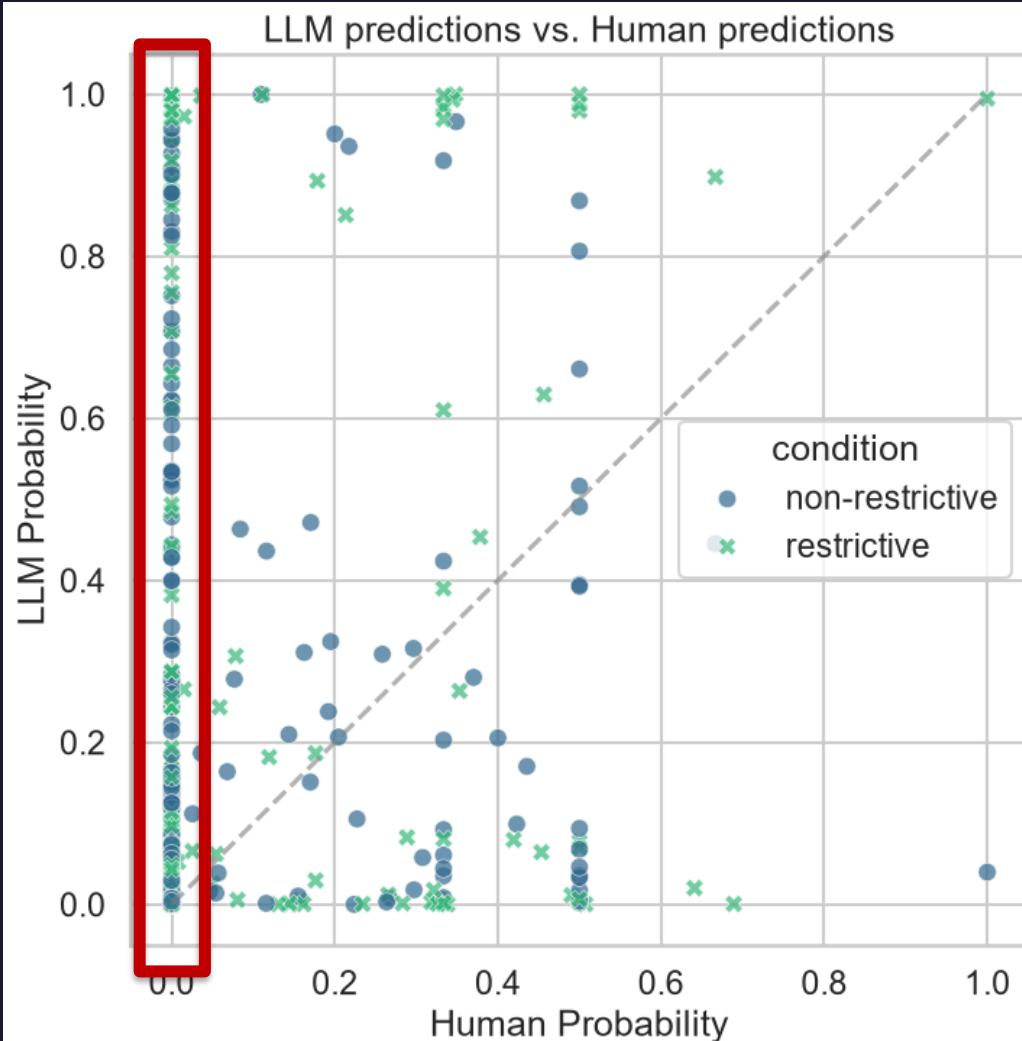
- Process the sentence incrementally at the same prediction window
- Obtain: probability distribution mapped to objects, surprisal, entropy

Compare Distributions

- Visualize relation in scatterplots
- Calculate JSD (Jensen–Shannon Divergence) per sentence
- (planned supportive analysis: Spearman Correlation)



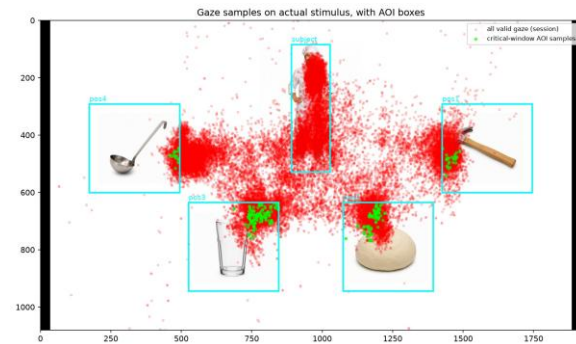
Preliminary results



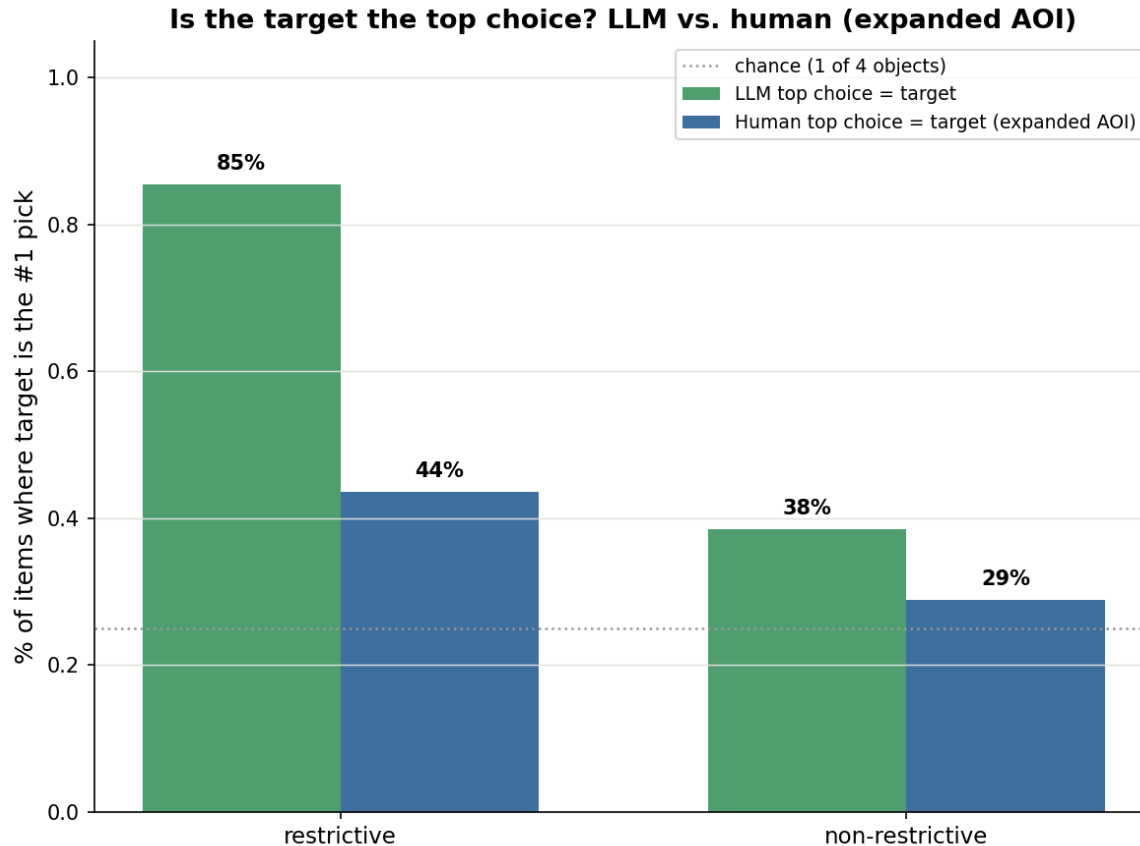
No clear correlation found

Reason:

participants are not looking at four AOI, but mostly still looking at the subject.
Critical window too short



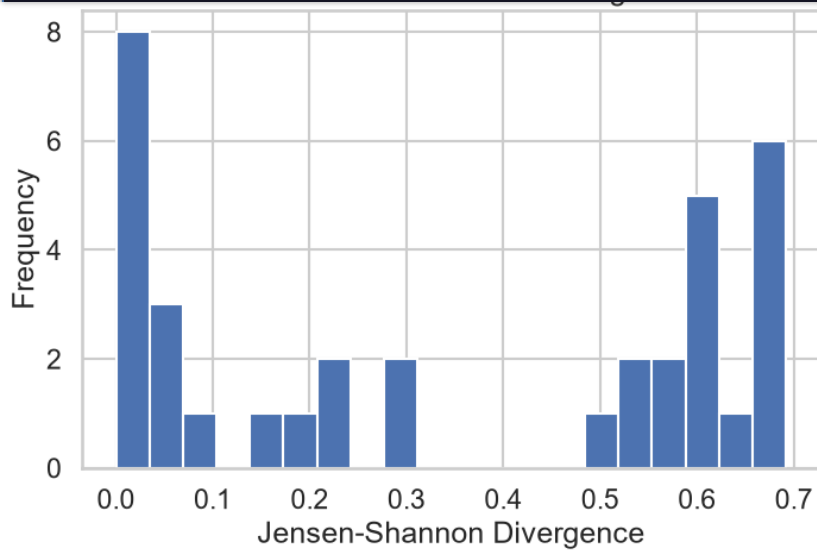
Preliminary results



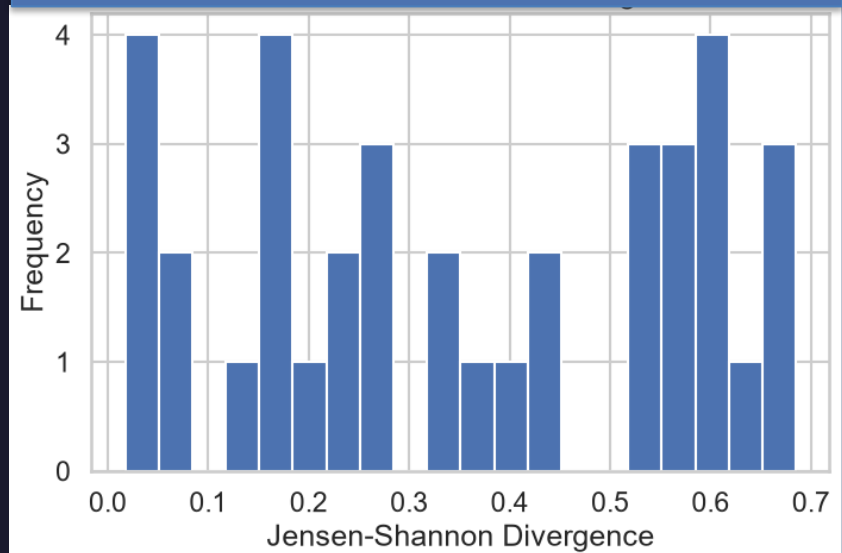
- Restrictive condition: LLM picks the target as its top choice **85%** of the time; humans pick it **44%** of the time, above chance (25%, since it's a 1-in-4 choice).
- Non-restrictive condition: humans pick the target only 29% of the time — essentially at chance level, as expected (a non-restrictive verb shouldn't provide any predictive cue).

Results II

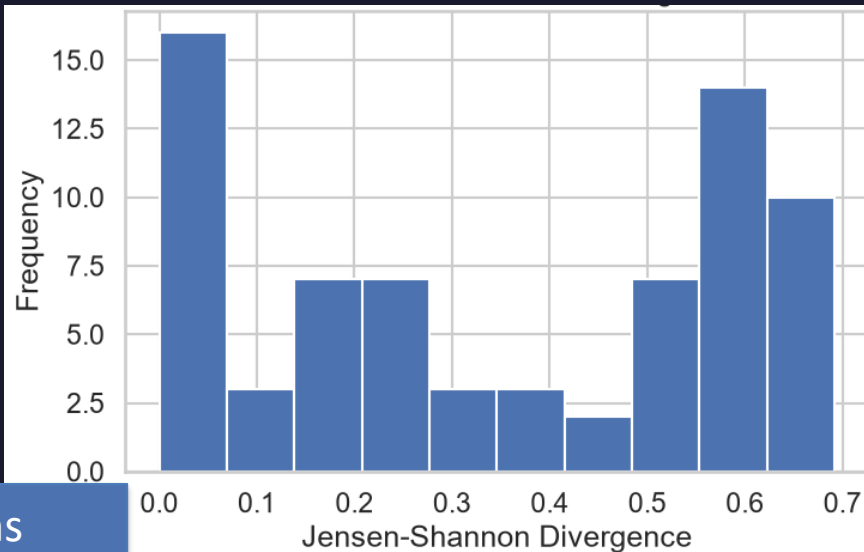
Restrictive-prediction only



Non-restrictive-prediction only



both conditions



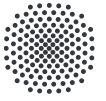
Further work & limitations

Further work

- collecting about 18 participants in total
→ get more telling prediction probabilities
- further analysis:
 - also samples that are part of saccades towards the AOI could be counted
 - other metrics, eg. percentage of equal top-target
- audio stimulus: even slower in future studies? (tradeoff: natural language)
- further sanity checking is warranted (annotation)

Limitations

- extension of sentence-dataset relies on its own assessment (restrictive/non-restrictive)
- biases from subject (doctor/cook) are introduced
- some objects come up more than once (eg envelope)



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Thank you